



ZIMBABWE

Ministry of Primary and Secondary Education



MATHEMATICS

SYLLABUS B

2024-2030

FORMS 1-4

Curriculum Development and Technical Services
Box MP 133
Mt Pleasant
Harare

@ 2024
All rights reserved

Acknowledgements

The Ministry of Primary and Secondary Education wishes to acknowledge the following for their valued contributions in the development of this syllabus:

- Learning area panellists for Form 1 to 4 Mathematics
- Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development (MoHTEISTD)
- Zimbabwe School Examinations Council (ZIMSEC)
- United Nations Children’s Fund (UNICEF)
- United Nations Educational Scientific and Cultural Organisation (UNESCO)

TABLE OF CONTENTS

Contents

Page

Table of Contents

Acknowledgement.....	ii
TABLE OF CONTENTS.....	iii
1.0 Preamble	1
1.1 Introduction	1
1.2 Rationale.....	1
1.3 Summary of Content	1
1.4 Assumptions.....	1
1.5 Cross Cutting themes	1
2.0 Presentation of Syllabus	2
3.0 Aims.....	2
4.0 Objectives.....	2
5.0 Topics	3
6.0 Methodology and Time Allocation	3
6.1 Time Allocation	4
7.0 SCOPE AND SEQUENCE CHART	4
8.0 CONTENT MATRIX: FORM 1 SYLLABUS.....	1
8.0 CONTENT MATRIX: FORM 2 SYLLABUS.....	17
8.0 CONTENT MATRIX: FORM 3 SYLLABUS.....	36
8.0 CONTENT MATRIX: FORM 4 SYLLABUS.....	54
9.0 Assessment	1
9.1 Assessment Objectives	Error! Bookmark not defined.
9.2 Assessment Model	81
9.3 Scheme of Assessment.....	Error! Bookmark not defined.

1.0 Preamble

1.1 Introduction

In developing the Heritage-based Mathematics syllabus B, attention was paid to the need to provide continuity of mathematical concepts from primary school level through to form 4. The syllabus is intended to empower learners with mathematical skills such as critical thinking and problem solving for use in solving everyday life problems. This four year learning course will provide learners with opportunities to integrate and apply mathematical concepts to other learning areas. It is also designed to inculcate into learners the ability to communicate effectively. Learners should understand and master various attitudes, skills, knowledge, concepts and processes in order to investigate and interpret numerical and spatial relationships and patterns that exist in the world.

1.2 Rationale

Mathematics is a fundamental component of economic education. It is very critical for driving innovation, entrepreneurship and industrialisation which are key to development. Therefore it is imperative that learners acquire necessary mathematical knowledge, skills, values and develop positive attitudes towards application of mathematical concepts to produce goods and services. This syllabus plays a pivotal role in careers such as engineering, construction, entrepreneurship, education, medicine, agriculture, meteorology and other trades.

1.3 Summary of Content

The Form 1 to 4 Heritage-based Mathematics syllabus B covers the theoretical and practical aspects of Mathematics. The content include simple mathematical operations and processes that can be applied in other learning areas especially the technical and vocational education and training categories.

1.4 Assumptions

The syllabus assumes that the learner has:

- grasped the primary mathematics concepts such as:
 - number
 - operations
 - measures
 - relationships
- acquired the ability to operate some IT gadgets

1.5 Cross Cutting themes

The following are some of the cross cutting themes in mathematics:-

- Business Enterprise Skills
- Disaster Risk Management
- Environmental Management
- Gender Equity
- Climate Change
- Health and wellbeing
- Child rights and responsibilities

2.0 Presentation of Syllabus

The Heritage-based Mathematics syllabus is a single document covering forms 1 to 4. It contains the preamble, aims, assessment objectives, syllabus topics, scope and sequence and content matrix. The syllabus also suggests a list of resources to be used during learning and teaching processes.

3.0 Aims

The syllabus is intended to provide a guideline for form 1 to 4 course that will enable learners to:

- 3.1 develop an understanding of mathematical concepts and processes in a way that encourages confidence, enjoyment and interest
- 3.2 further acquire appropriate mathematical skills and knowledge
- 3.3 develop the ability to think clearly, work carefully and communicate mathematical ideas successfully
- 3.4 apply mathematics in other learning areas and in life
- 3.5 develop an appreciation of the role of mathematics in personal, community and national development
- 3.6 engage, persevere, collaborate and show intellectual honesty in performing tasks in mathematics, in the spirit of Unhu/ Ubuntu/Vumunhu

4.0 Objectives

The learners should be able to:

- 4.1 use mathematical symbols, terms and definitions in problem solving
- 4.2 construct appropriate mathematical models that can be applied in solving problems in life
- 4.3 draw inferences through manipulation of mathematical data
- 4.4 communicate mathematical ideas and information clearly and effectively in various contexts
- 4.5 solve a wide range of problems involving algebraic and geometric concepts
- 4.6 apply mathematical concepts in other learning areas
- 4.7 use I.C.T tools in solving Mathematical problem
- 4.8 conduct Mathematical research projects including those related to enterprise

5.0 Topics

The following topics will be covered from Form 1 to 4:

- Real Numbers
- Sets
- Money and Finance
- Measures and Mensuration
- Graphs
- Variation
- Algebra
- Geometry
- Statistics
- Trigonometry
- Vectors
- Matrices
- Probability
- Transformations

6.0 METHODOLOGY AND TIME ALLOCATION

6.1 Methodology

It is recommended that teachers use teaching techniques in which mathematics is seen as a process which arouse an interest and confidence in tackling problems both in familiar and unfamiliar contexts. The teaching and learning of mathematics must be learner centred. Ortho-didactic principles should also be applied during teaching and learning of mathematics. The following are suggested methods of the teaching and learning of mathematics

- Guided discovery
- Group work
- Interactive e-learning
- Exposition
- Demonstration and illustration
- Problem solving
- Individualisation
- Stimulation
- Self-activity
- Totality and wholeness
- Visual tactile
- Educational tours

6.2 Time Allocation

Six periods of 40 minutes each per week should be allocated for the adequate coverage of the syllabus

7.0 SCOPE AND SEQUENCE CHART

7.1 TOPIC 1: REAL NUMBERS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Number Concepts and Operations	• Number types • Factors and multiples • Directed numbers • Fractions and percentages • Order of operations	• Factors and multiples • Squares and square roots • Cubes and cube roots	• Order of operations • Irrational numbers • Number patterns	
Approximations and estimations	• Round off numbers • Decimal places	• Significant figures • Estimations	• Estimations • Significant figures • Basic limits of accuracy	• Limits of accuracy
Ratios, rates and proportions	• Ratios	• Ratios • Proportions	• Ratios • Rates • Proportions	
Ordinary and standard form	• Large and small numbers	• Numbers in standard form	• Operations in standard form	
Number bases	• Number bases in everyday life • Place values	• Converting numbers from one base to another	• Operations in number bases	
Scales and simple map problems	• Scale measurement	• Scale drawing	• Scales • Length factor • Area factor	

TOPIC 2: SETS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Sets	- Sets and Set notation - Types of sets	- Types of sets - Venn diagram with two subsets	- Set Builder Notation - Venn diagrams with three subsets	

TOPIC 3: FINANCIAL MATHEMATICS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Consumer arithmetic	- Household bills - Profit and loss - Discount - Household budgets	- Corporate bills - Profit and loss - Simple interest - Hire purchase - Small scale enterprise budgets	- Bank statements - Compound interest - Commission - Hire purchase	- Foreign exchange - Sales and income tax rates (Pay as you earn (PAYE)) - Value added tax (VAT) - Customs and excise duty

TOPIC 4: MEASURES AND MENSURATION

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Measures	Units of : - Time - Mass - Length - Temperature - Capacity	Units of: - Area - Volume - Capacity - Density		
Mensuration	- Perimeter of plane shapes - Area of plane shapes	Perimeter of plane shapes	Perimeter of combined shapes	Area and volumes of similar shapes

	Volume of cuboids	- Area of plane shapes - Volume of cylinders	- Area of combined shapes - Volume of cones and pyramids - Density	- Surface area - Volume of prisms
--	---	---	--	--

TOPIC 5: GRAPHS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Functional Graphs	- Cartesian plane - Scale - Co-ordinates	- Cartesian plane - Table of values - Linear graphs - Scale	- Functional Notation - Linear graphs - Quadratic graphs	- Cubic graph - Inverse graph
Travel Graphs	Distance time graphs	Distance time graphs	- Distance time graphs - Speed-time graphs	- Displacement time graphs - Velocity-time graphs

TOPIC 6: VARIATION

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Variation		Direct variation	- Direct variation - Inverse variation	- Joint variation - Partial variation

TOPIC 7: ALGEBRA

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Algebraic Manipulation	- Basic arithmetic processes in letter symbols - Substitution of values	- Substitution of values - Algebraic expressions - Algebraic fractions	- Algebraic fractions - Highest Common Factor (HCF) and Lowest	- Algebraic fractions - Quadratic expressions - Factorisation

	Algebraic expressions	- Quadratic expressions - Factorisation	Common Multiple (LCM) of algebraic expressions - Quadratic expressions - Factorisation	Completing the square
Equations	Linear equations	- Equations with brackets - Equations with fractions - Change of subject of formulae - Simultaneous linear equations - Quadratic equations	- Simultaneous equations - Quadratic equations - Change of subject in formulae - Substitution of values	- Completing the square - Quadratic formulae
Inequalities	Linear inequalities	- Linear inequalities - Number line - Cartesian plane	- Simultaneous inequalities - Graphs of inequalities	Linear programming
Indices and Logarithms	Index form	- Laws of indices - Highest Common Factor (HCM) and Lowest Common Multiple (LCM) in index form - Squares and square roots	- Indices - Logarithms - Theory of logarithms - Equations involving indices and logarithms	

		• Cubes and cube roots		
--	--	--	--	--

TOPIC 8: GEOMETRY

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Points, lines and angles	• Points • Lines • Angles	• Angles, • Transversal and parallel lines •	• Angles of elevation and depression	
Bearing		• Cardinal points • Three figure bearing • Compass bearing	• Three figure bearing • Compass bearing	
Polygons	• Polygons	• Properties of polygons (triangles and quadrilaterals)	• Properties of polygons (pentagon to circle) • Angles properties of polygons • Circle theorems • Numbers of sides of polygons	• Circle theorems
Similarity and Congruency		• Similar and congruent figures • Similar triangles • Congruent triangles	• Areas of similar figures • Volume and mass of similar solids	
Constructions and Loci	• Construction of lines and angles	• Construction of angles • Bisecting lines and angles	• Construction of triangles and quadrilaterals	• Construction of diagram to a scale • Define loci

				Construction of loci
Symmetry	Line symmetry in two dimensions	Line symmetry in two dimensions	Rotational symmetry in two dimensions	

TOPIC 9: STATISTICS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Data Collection and Classification	- Data collection - Data classification	- Data collection - Classification of ungrouped data	Collection and classification of grouped data	
Data Representation	- Pictograms - Bar chart - Frequency distribution table	- Bar chart - Frequency distribution table - Pie chart	- Frequency table - Bar chart - Pie chart - Histogram - Frequency polygon	- Frequency table - Histogram - Frequency polygon - Cumulative frequency table - Cumulative frequency curve
Measures of Central Tendency		- Mean - Mode - Median	- Mean, median and mode of grouped data - Assumed mean	Median from cumulative frequency curve
Measures of Dispersion				- Quartiles - Inter quartile range

Type equation here.

TOPIC 10: TRIGONOMETRY

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Pythagoras Theorem			• Introduction to Pythagoras theorem	
Trigonometrical Ratios			• Trigonometrical ratios of acute angles: - Sine - Cosine - Tangent • Trigonometrical ratios of obtuse angles	• Cosine rule • Sine rule

TOPIC 11: MATRICES

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Dimension/ Order		• Order/Dimension of matrices • Types of matrices		
Operations		• Addition and subtraction of matrices	• Addition and Subtraction of matrices • Scalar multiplication of matrices • Multiplication of matrices	
Determinants			• Determinants of Matrices • Singular and non-singular matrices	
Inverse Matrix			• Inverse of matrix • Simultaneous linear equations in 2 variables	•

TOPIC 12: VECTORS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Definition and Notation		- Definition of vector - Vector notation		
Types of Vectors		- Translation vectors - Negative vectors - Equal vectors - Parallel vectors	- Translation - Negative - Equal - Parallel - Position	
Operation		- Addition of vectors - Subtraction of vectors	- Addition of vectors - Subtraction of vectors - scalar multiplication - Magnitude - Combined vector operations	Vector properties of plane shapes

TOPIC 13: TRANSFORMATIONS

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Translation	Translation of plane figures	Translation vector to move a point	translation vector to move a plane figure on Cartesian plane	use of translation vectors
Reflection		Reflection of plane figures	Reflection of plane figures on a Cartesian plane in the x-axis, y-axis, lines of the form $y=a$ and $x=b$	Reflection of plane figures in any line and use of matrices
Rotation			Rotation of plane figures in a	Rotation of plane figures by drawing

			Cartesian plane by drawing	and use of matrices
Enlargement			Enlargement about the origin using a rational scale by drawing	Enlargement using matrices and about any point using a rational scale
Stretch				One-way and two-way stretch using matrices and geometrical methods
Shear				Shear using matrices and geometrical methods

TOPIC 14: PROBABILITY

TOPIC	FORM 1	FORM 2	FORM 3	FORM 4
Probability		- Definition of probability terms - Experimental probability	- Experimental probability - Theoretical probability - Single events	- Combined events - Application of probability

FORM ONE (1)

8.1 REAL NUMBERS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Number Concepts and operations	- identify types of numbers - find factors and multiples of numbers - find H.C.F. and L.C.M. - operate with directed numbers	- Number types - Factors and multiples - Highest Common Factor (H.C.F.) and Lowest Common Multiple (L.C.M.) - Directed numbers	- Identifying and listing types of numbers - Listing factors and multiples of numbers - Finding H.C.F and L.C.M. - Using a number line on the operation of directed numbers - Using counters such as bottle tops	- Counters such as Bottle tops - Number line - Approved and Recommended Textbooks - Charts with relevant concepts

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	• apply directed numbers to practical situations in everyday life • operate with fractions • convert fractions to decimals • convert fractions to percentages and vice versa • carry out calculations involving percentages	• Fractions • Decimals • Percentages	on the operation of directed numbers • Operations with fractions • Converting fractions to decimals • Converting fractions to percentages • Calculations involving decimals and percentages	
Approximation and Estimation	• round off numbers to the given place value • round off numbers to the given decimal places	• Whole numbers • Decimal numbers	• Rounding off numbers <i>N.B. Digits 0;1;2;3;4; are rounded down and digits 5;6;7;8;9 are rounded up</i>	• Approved and Recommended Textbooks • calculators and computers

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Ratios	- simplify ratios - solve problems involving ratios	Ratio	- Simplifying ratios - Solving problems using ratios	Approved and Recommended Textbooks
Large and small numbers	express small and large numbers in digits and words or vice versa	Large and small numbers	expressing small and large numbers in digits and words or vice versa	- Approved and recommended textbooks - Calculator
Number bases	- identify number bases in everyday life situations - identify numbers in their respective bases - identify place value of numbers	- Time related bases such as seconds, minutes, hours and weeks - Quantity related bases such as dozen and score - Language related bases - Place value	- Identifying different number bases - Identifying numbers in their respective bases - Identifying place value	- Approved and Recommended Textbooks - Charts - Computer

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Scales	- Identifying types of scales - find scales from given information - measure lengths using a given scale - use given scales to draw lines or diagrams - calculate distances using a given scale	- Representative Fraction - Ratio scale	- Identifying types of scales - Measuring lengths using given scales - Making scale drawings using appropriate/given scale - Calculating distances	- Plans - Maps - Geometrical instruments - Approved and Recommended Textbooks

8.2 SETS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Sets and Set notation	- define a set by listing the elements - define a set by describing common characteristics of elements	- Sets - Elements - Number of elements in a set	- Listing elements of various sets - Describing the common characteristics of elements	- Mathematical set - Sets of objects - Charts

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- state the number of elements in a set - name and match set symbols		Naming and matching symbols and notation of sets for example: $A = \{ \dots \}$ for a set ϵ: for element of a set $\notin$: for not an element of $n(A)$: for number of element in set A	
Types of Sets	- state the types of sets - form subsets	- Universal set - Finite set - Infinite set - Null or empty set - Equal sets - Subset - Union of a set - Intersection of a set	- Listing of types of sets - Identify types of sets - Dividing elements of universal set into subsets - Connecting and matching sets	- Charts with various sets - Cutlery set - Mathematical set - Empty box - Approved and Recommended Textbooks

8.3 FINANCIAL MATHEMATICS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Consumer arithmetic	• Interpret bills • Extract data from household bills for calculations • calculate profit and loss • calculate discount • prepare household budgets	• Household bills • Buying and selling • Profit and loss • Discount • Household budgets	• Interpreting household bills and corporate bills • Solving problems involving household bills • Calculating profit and loss • Calculating discount • Preparing and discussing household budget	• Relevant Textbooks • Corporate and household bills • Resource persons • Newspaper cuttings

8.4 MEASURES AND MENSURATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Measures	• read time on both the 12 and 24 hour clock • identify SI units of quantities • express one quantity in terms of smaller or larger units • measure different quantities	• Time • Mass • length • Temperature • Capacity	• Reading time • Expressing quantities in terms of smaller or larger units • Naming SI units of quantities • Measuring of quantities	• Watches • Clocks • Scales • Thermometer • Mathematical instruments • Approved and Recommended Textbooks • Measuring instruments
Mensuration	• calculate perimeter of plane shapes • calculate area of plane shapes • calculate volumes of solid shapes	• Plane shapes such as: Rectangles, circles, triangle, parallelogram, trapezium • Solid shapes such as cuboids and prisms	• Calculating perimeter and area of plane shapes • Calculating volume of solid shapes	• Approved and Recommended Textbooks • Plane shapes • Cuboids such as chalk boxes • Prisms

8.5 GRAPHS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Functional graphs	- draw the Cartesian plane using a given scale - name points on the Cartesian plane - plot points on the Cartesian plane	- Cartesian plane - Scale	- Naming co-ordinates of points on the Cartesian plane - Plotting points on the Cartesian plane	- Graph books - Approved and Recommended Textbooks - Geo-board - Mathematical instruments
Travel graphs	interpret distance time graphs	Distance time graphs	Solving problems on distance time graphs	- Distance time graphs - Approved and Recommended Textbooks

8.6. ALGEBRA

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Symbolic Expression	represent unknown quantities with letters/symbols	Algebraic expressions	Expressing given information in letters	Approved and Recommended Textbooks
Algebraic Manipulation	- group like terms - simplify using the rules for the four basic operations - substitute values for letters - find H.C.F of linear algebraic expressions - factorise linear algebraic expressions	- Monomials - Algebraic expressions with two or more variables - Substitution - Factorisation of linear algebraic expressions	- Grouping terms - Simplifying algebraic expressions using the rules of the four operations - Substituting values for letters - finding H.C.F of linear algebraic expressions - Factorising linear algebraic expressions	Approved and Recommended Textbooks
Equations	- solve linear equations where the unknown appears one side - solve linear equations where the unknown appears	Linear equations	- Solving linear equations where the unknown appears on one side - Solving linear equations where	- Approved and Recommended Textbooks - Work cards

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- on both sides of the equation - formulate linear equations from given information	Word problems	- the unknown appear on both sides - Forming linear equations from given information	
Inequalities	- define the words at least, at most, less than, more than using inequality signs - represent linear inequalities on a number line - formulate linear inequalities - solve linear inequalities	- Inequality signs - Linear inequalities	- Defining the terms at least, at most, less than and greater than - Representing linear inequalities on a number line - Formulating linear inequalities - Solving linear inequalities	- Charts with inequalities - Approved and Recommended Textbooks
Algebraic expressions in index form	express product of letters in index form	- Index form - Common factors - Common multiples	Expressing the product of letters in index form	Approved and Recommended Textbooks

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- find common factor of algebraic expressions - find common multiples of algebraic terms - find highest common factor and lowest common multiple of algebraic expressions	- Highest common factor - Lowest common multiples	- Finding common factors of algebraic expressions - Finding common multiples of algebraic terms - Finding highest common factors of algebraic expressions	

8.7. GEOMETRY

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Lines and Angles	- Identify type of lines - define an angle - draw angles - measure angles - Identify types of angles	- Lines - Angles	- Identifying types of lines - defining an angle - drawing angles - measuring angles	- Geometrical Instruments - Approved and Recommended Textbooks

	- calculate angles on a straight line and at a point - identify vertically opposite angles - identify angles formed on parallel lines cut by transversal lines - calculate angles on parallel lines cut by transversal lines	- Acute , right angle, obtuse, straight line and reflex angles - Angles on a straight line or at a point - Vertically opposite angles - Alternate angles - Corresponding angles - Allied/co-interior angles	- Identifying types of angles - Calculating angles on a straight line and at a point - Identifying vertically opposite angles - Naming angles on parallel lines cut by transversal lines - Calculating angles on parallel lines cut by transversal lines	
Polygons	- define a polygon - state the names of n-sided polygons (up to $n=10$)	Polygons	Naming polygons with up to ten sides	- Approved and recommended textbooks - Polygonal shapes - ICT tools
Circles	name parts, lines and regions in a circle	Parts of a circle	Drawing and naming parts of a circle	- Approved and recommended textbooks - ICT tools

				Mathematical instruments
Construction	construct lines and angles	- Lines - Angles	Constructing lines and angles	- Geometrical instruments - ICT tools - Approved and recommended textbooks
Symmetry	identify lines of symmetry for regular polygons	- plane shapes with line symmetry - Isosceles triangles - Equilateral triangle - Regular polygons - Parallelograms - Letters with lines of symmetry - Circle	- Stating number lines of symmetry - Drawing lines of symmetry	- Approved and recommended textbooks - Shapes - Letters - Regular polygons

8.8. STATISTICS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Data collection and classification	- collect statistical data - classify statistical data - tabulate statistical data	- Statistical data - Frequency table	- Collecting statistical data - Classifying statistical data - Tabulating statistical data - Case study - Administering a questionnaire	- Newspapers - Archival information - Population samples - Resource person - ICT tools - Questionnaire
Data representation	- tabulate statistical data - represent data on bar chart, pie chart - interpret a pictogram	- Bar chart - Pie chart - Pictogram - Frequency table	- Tabulating statistical data - Representing statistical data on bar chart and pie chart - Interpreting pictogram - Taking educational tour	Approved and recommended textbooks

8.9. TRANSFORMATIONS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Translation	• define transformation • define translation • translate plane figures	• Translation	• Defining transformation • Translating plane figures	• Approved and recommended textbooks • Geo-board • Flannel board • Charts • ICT tools

FORM TWO (2)

8.0 COMPETENCY MATRIX

8.1 REAL NUMBERS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Number Concepts and operations	- find factors and multiples of numbers - find H.C.F. and L.C.M. - find squares and square roots - find cubes and cube roots	- Factors and multiples - Highest Common Factor (H.C.F.) and Lowest Common Multiple (L.C.M.) - Squares and square roots - Cubes and cube roots	- Listing factors and multiples of numbers - Finding H.C.F and L.C.M. - Finding squares and square roots - Finding cubes and cube roots	- Approved and Recommended Textbooks - Charts with relevant concepts - ICT tools
Approximation and Estimation	- Round off numbers to given significant figures - Estimate to a given degree of accuracy	Significant figures	Rounding off numbers to required significant figures <i>N.B. Digits 0;1;2;3;4; are rounded down</i>	- Approved and Recommended Textbooks - ICT tools

		Estimation	and digits 5;6;7;8;9 are rounded up	
Ratios and Proportions	- simplify ratios - use the idea of ratio to solve problems - use direct and inverse proportion to solve problems	- Ratio - Direct proportion - Inverse proportion	- Simplifying ratios - Solving problems using ratios or proportion	Approved and Recommended Textbooks
Standard Form	express numbers in standard form	Scientific notation	Expressing small and large numbers in the form: $A \times 10^n$ where $1 \leq A < 10$ and n is an integer	- Approved and recommended textbooks - ICT tools
Number bases	- convert any number to base ten - convert a number in base ten to any base	- Bases: 2;3;4;5;6;7;8;9;10 - Changing number bases	- Expanding number bases - Converting number bases - Identifying numbers in their respective bases	- Approved and Recommended Textbooks - ICT tools
Scales	- identify types of scales - find scales from given information - measure lengths using a given scale	- Representative Fraction - Ratio scale	- Identifying types of scales - Measuring lengths using given scales - Making scale drawings using	- Plans - Maps - Geometrical instruments - Approved and Recommended Textbooks

	- draw lines or diagrams to a given scale - calculate distances using a given scale	Scale drawings	- appropriate/given scale - Calculating distances	
--	--	--	--	--

8.2. SETS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Types of Sets	find union and intersection of sets	- Universal set - Union of a set - Intersection of a set	Finding union and intersection of sets	- Charts with various sets - Mathematical sets - ICT tools
Venn diagrams	- represent sets on Venn diagrams - solve life problems using Venn diagram with no more than 2 subsets - convert word problems into set notations	Venn diagrams with no more than 2 subsets	- Using information on Venn diagram to solve problems - Converting word problems to set notation - Representing information on a Venn diagram	Approved and recommended textbooks

8.3. FINANCIAL MATHEMATICS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Bills	• Interpret bills • Use information from a bill • Extract data from household bills for calculations	• Household bills • Corporate bills	• Interpreting household bills and corporate bills • Extracting data from household bills for calculations	• Approved and recommended textbooks • Sample of household bills • Resources persons • Print media
Consumer arithmetic	• calculate profit and loss • calculate discount • find simple interest • solve problems on hire purchase • prepare an enterprise budget for a small business	• Buying and selling • Simple interest • Hire purchase • Enterprise budget	• Calculating profit loss and discount • Finding simple interest • Solving problems on hire purchase • Preparing an enterprise budget for a small business	• documents in trade • information from media • Approved and recommended textbooks • Business financial documents

8.4. MEASURES AND MENSURATION

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Mensuration	• calculate perimeter of plane shapes • calculate area of plane shapes • calculate volumes of shapes	• Perimeter of plane shapes • Area of plane shapes • Volume of solid shapes	• Calculating perimeter and area of plane shapes • Calculating volume of solid shapes	• Approved and recommended textbooks • Plane shapes • Cuboids • ICT tools • Cylinders

8.5. GRAPHS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Functional graphs	- draw the Cartesian plane, using a given scale - plot points on the Cartesian plane - construct a table of values for a given linear function - draw a straight line graphs	- Cartesian plane - Scale - Table of values - Straight line graphs	- Drawing the Cartesian plane, using a given scale - Plotting points on the Cartesian plane - Constructing table of values - Drawing straight line graphs on the Cartesian plane	- Graph books - Approved and recommended textbooks - Mathematical instruments - ICT tools (Fluid Math Software)
Travel graphs	- interpret distance time graphs - draw distance time graphs - use distance-time graphs to answer questions	Distance time graphs	- Interpreting distance time graphs - Solving problems on distance time graphs	- Graph books - Distance time graphs - Approved and recommended textbooks

			• Drawing distance time graphs • Using distance-time graphs to answer questions	
--	--	--	--	--

8.6. VARIATION

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Direct Variation	• express direct variation in algebraic terms • solve problems in direct variation • read and interpret a ready reckoner	• Direct variation in algebraic terms	• Expressing a statement in terms of an unknown constant • Finding the unknown constant • Calculating one of the variation using the constant • Finding relationships between variables • Reading and interpreting ready reckoner	• Approved and recommended textbooks • Extract from newspapers • Ready reckoner • ICT tools (GPS)

Variation Graphs	- draw direct variation graphs - interpret the information on given graphs	Variation graphs	- Drawing graphs - Interpreting information on graphs - Copy and complete tables	- Approved and recommended textbooks - Graph papers
------------------	---	--	--	--

8.7. ALGEBRA

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Algebraic Manipulation	- substitute values for letters - factorise linear algebraic expressions - factorise quadratic algebraic expressions - simplify algebraic fractions - find Highest Common Factor and	- Algebraic expressions with two or more variables - Linear expressions - Quadratic expressions - Algebraic fractions	- Substituting values for letters - Factorising linear and quadratic algebraic expressions - Simplifying algebraic fractions - Finding H.C.F. and L.C.M. of algebraic expressions - Expanding algebraic expressions with brackets	Approved and recommended textbooks

	Lowest Common Multiple of algebraic expressions • expand algebraic expressions with brackets	• H.C.F and L.C.M • Algebraic expressions with brackets		
Equations	• solve linear equations where the unknown appears on both sides of the equation • construct linear equations from given information • solve equations with brackets • solve equations with algebraic fractions • solve simultaneous linear equations • solve quadratic equation where the coefficient of x^2 is one	• Linear equations • Equations with brackets • Equations with fractions • Simultaneous equations	• Solve linear equations where the unknown appears on both sides • Constructing linear equations from given information • Expanding and solving equations with brackets • Simplifying and solving algebraic fractions • Solving simultaneous linear equations • Solving quadratic equations where the	• Approved and recommended textbooks • Work cards

		Quadratic equations	coefficient of x^2 is one	
Inequalities	- represent linear inequalities on a number line - formulate linear inequalities - solve linear inequalities - represent inequalities on a Cartesian plane	linear inequalities	- Representing linear inequalities on a number line - Constructing linear inequalities - Solving linear inequalities - Drawing vertical and horizontal lines with respect to the inequality signs - Shading the unwanted region	- Charts - Approved and recommended textbooks - ICT tools
Indices	- apply the laws of indices - calculate squares and square roots of algebraic expressions - calculate cube and cube roots of algebraic expressions	- Laws of indices - Squares and square roots - Cube and cube roots.	- Applying the laws of indices to algebraic expressions NB: $x^a \times x^b = x^{a+b}$ $x^a \div x^b = x^{a-b}$ $x^0 = 1$ - Calculating squares and square roots of algebraic expressions	Approved and recommended textbooks

			Calculating cube and cube roots of algebraic expressions	
--	--	--	--	--

8.8. GEOMETRY

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Bearings	- name cardinal points - give directions using cardinal points - find compass bearings of points - calculate three-figure bearings of points	- Cardinal points - Compass bearings - Three-figure bearings	- Naming cardinal points - Finding compass bearings - Calculating three-figure bearings	- Magnetic compass - Approved and recommended textbooks - ICT tools
Polygons	- state the names of n-sided polygons (up to $n=10$) - give properties of triangles and quadrilaterals	Polygons	- Naming polygons with up to ten sides - Stating properties of triangles and quadrilaterals	- Approved and recommended textbooks - Polygons - ICT tools

Similarity and congruency	• identify similar and congruent figures • solve problems on similar and congruent figures	• Similar and congruent triangles	• Identifying similar and congruent figures	• Approved and recommended textbooks • ICT tools • Similar and congruent figures
Construction	• construct lines and angles • bisect lines and angles	• Lines • Angles	• Constructing lines and angles • Bisecting lines and angles	• Geometrical instruments • ICT tools • Approved and recommended textbooks
Symmetry	• identify lines of symmetry for regular polygons	• Isosceles triangles • Equilateral triangle • Regular polygons • Parallelograms • Letters with lines of symmetry • Circle	• Stating number lines of symmetry • Drawing lines of symmetry	• Approved and recommended textbooks • Shapes • Letters • Regular polygons

8. 9. STATISTICS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Data representation	- group statistical data - represent data on bar chart, pie chart	- Bar chart - Pie chart	- Grouping statistical data - Representing statistical data on bar chart and pie chart - Taking a educational tours	- Approved and recommended textbooks - ICT tools
Measures of central tendency	- define measures of central tendency - state the mode - calculate the median - calculate the mean	Measures of central tendency	- Demonstrating how to get the mode and median - Stating the mode - Calculating the mean and median	- Mark schedules - Sample population - Approved and recommended textbooks
Measures of dispersion	- define the concept of range in statistics - calculate range	Range	- Defining range - Demonstrating how to find range - Calculating range	- Approved and recommended textbooks - ICT tools

8.10. MATRICES

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Dimension and order	- define a matrix - state the order/dimension of a given matrix - name different types of matrices	- Column (2x1) matrix - Row matrix (1x2) - Square matrix (2x2)	- Recording information in matrix form - Stating the order/dimension - Locate elements in a given matrix - Listing types of matrices	- Approved and recommended textbooks - Cashbox with compartments
Operations	- Adding Matrices - Subtracting matrices	- column matrices (2x1) - Row matrices (1x2) - Square matrices (2x2)	- Identifying matrices of the same order - Adding or subtracting corresponding elements	Approved and recommended textbooks

8.11 VECTORS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Definition of a vector	- define vector - use vector notation properly	- Magnitudes - Direction - Examples of vectors a	- Listing vectors in column form - Representing vectors using vector notation	Approved and recommended textbooks
Types of vectors	- identify various types of vectors - use the concepts of various types of vectors	- Translation - Negative vector - Equal vector - Parallel vector	- Representing a translation by a column vector - Drawing translation vector on a Cartesian plane - Identifying various types of vector from the Cartesian plane	- Graph paper - Approved and recommended textbooks

Operations	• add or subtract two or more vectors	• Addition • Subtraction of vectors	• Adding or subtracting vectors	• Approved and recommended textbooks
------------	---	--	---	--

8.12 TRANSFORMATIONS

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Translation	• define transformation • define translation • translate plane figures and points	• Translation • Reflection	• Identifying transformations • Translating points using translation vector (T) <i>N.B: Object coordinates + Translation vector (T) = Image coordinates</i> • Translating plane figures	• Approved and recommended textbooks • Graph papers • Charts • ICT tools

Reflection	- define reflection - reflect a plane figure in a given mirror line	Reflection	Reflecting a point or object in a mirror line	- Graph papers - Approved and recommended textbooks - ICT tools
------------	--	--	---	---

8.12 PROBABILITY

TOPIC	OBJECTIVES LEARNERS SHOULD BE ABLE TO:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Definition of probability	- define probability - define probability terms	- Probability - Probability Terms	Stating examples of each probability term	- Approved and recommended textbooks - Counters - Playing cards - Dice - Coins - Darts - ICT tools

Experimental probability	• describe experimental probability	• Experimental probability	• Calculating probability of single events • Carry out probability experiments with relevant resources	• Approved and recommended textbooks • Counters • Playing cards • ICT tools • Dice • Coins • Darts
--------------------------	---	--	---	--

FORM THREE

MATHEMATICS SYLLABUS E1-4 2024-2030

8.0 COMPETENCY MATRIX

TOPIC 1: REAL NUMBERS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Number Concepts and operations	- perform arithmetic calculations using order of operations - identify rational and irrational numbers - distinguish between rational and irrational numbers - perform calculations involving irrational numbers - identify the sequence of number patterns	- Order of operations: - Natural numbers - Common/decimal fractions - Percentages - Integers - Surds - Directed numbers - Rational and irrational numbers - Patterns: - Numbers - Shapes	- Applying the rules of precedence in real numbers - Rule of precedence; bracket first, multiplication and division - Addition and subtraction - Differentiating between rational and irrational numbers - Simplifying problems involving irrational numbers - Exploring and discovering number patterns	- Number line - Objects in the environment
Approximation and Estimation	- approximate to the given degree of accuracy - state the numbers of significant figures in given numbers - round off numbers to given significant figures - identify the limits of accuracy - make estimates of quantities	- Estimations - Significant figures - Basic limits of accuracy - Decimal places - Approximations	- Rounding off numbers to given degree of accuracy - Stating the numbers of significant figures in a given number - Finding the lower limit and upper limit and use them to simplify problems	- Newspaper cuttings - Maps - Local environment

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	distinguish between approximation and estimation		Discussing the differences between approximation and estimation	
Ratios, rates and Proportions	- simplify ratios - perform calculations using ratios - solve problems involving rate - apply direct and inverse proportion to solve problems	- Ratio - Rates - Proportion - Direct - Inverse	- Reducing ratios to simplest form - Sharing quantities using ratio - Calculating using rates - Solving problems involving direct and inverse proportion focussing on life situations	- Newspaper cuttings - Objects
Ordinary and standard form	perform operations in standard form	Operations in standard form	- Adding and subtracting numbers in standard form - Dividing and multiplying in standard form	- Newspaper cuttings - Magazines
Number bases	perform operations in bases	- Operations in number bases - Bases 2,3,4,5,6,7,8,9 &10	- Adding and subtracting in number bases - Solving equations involving number bases	- Objects - Calendar - Clock
Scales and simple map problems	use given scales to calculate distances and areas	- Scales - Scale factor - Area factor	- Calculating distances and areas - Finding area factor given the scale factor - Finding scale factor given the area factor - Applying scales in solving problems in life situations	Maps

TOPIC 2: SETS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Set Builder Notation	define a set using a set builder notation	Symbols in sets	- Listing elements of sets - Using symbols of sets to describe sets	Objects in the local environment
Venn diagrams	- draw Venn diagrams to show relationships in different subsets - use Venn diagrams to solve problems in sets	Venn diagrams	- Demonstrating relationships of different subsets - Discussing Venn diagrams with three subsets - Solving problems involving Venn diagrams	Mathematical sets

TOPIC 3: FINANCIAL MATHEMATICS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Consumer arithmetic	- interpret bank statements - calculate compound interest - calculate commission - solve problems on hire purchase	- Bank statements - Compound interest - Commission - Hire purchase	- Discussing the contents of the bank statements - Extracting data from bank statements to use it for calculations - Discussing compound interest, commission and hire purchase - Computing compound interest, commission and hire purchase	- Newspaper cuttings showing statements - Bank statements - Invoice statements - Hire purchase statements

TOPIC 4: MEASURES AND MENSURATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Mensuration	- calculate perimeter of combined shapes - calculate area of combined shapes	- Perimeter of combined shapes - Area of combined shapes	- Discussing perimeter and area of combined shapes - Discussing volume of cone and pyramid	- ICT Tools - Relevant text - Models of solid shapes

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- calculate volume of cone and pyramid - calculate density of objects	- Volume of cones and pyramids - Density	- Carrying out an experiment to show the relationship between mass and volume - Solving problems involving mass, volume and density	

TOPIC 5: GRAPHS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Functional Graphs	- use the functional notation $f(x)$ in: - evaluating functions - solving linear and quadratic equations - draw graphs of linear and quadratic functions using: - table of values - axes of intercepts - sketch: - straight line - quadratic graphs using axes intercepts - use graphs to read off unknowns	- Functional notation - Linear graphs - Quadratic graphs	- Discussing use of functional notation using familiar functions e.g. perimeter of square/circle, area of square/circle - Drawing linear and quadratic graphs using - Tables of values - Axes intercepts - Sketching straight line and quadratic graphs using - Tables of values - Axes intercepts - Finding unknown values using: - Linear - Quadratic graphs	- ICT Tools - Relevant text
Travel Graphs	- relate distance, speed and time - draw distance-time graphs - draw speed-time graphs - solve problems involving travel graphs	- Speed - Acceleration - Distance-time graphs - Speed-time graphs	- Discussing relationship involving distance, speed and time in everyday life - Drawing distance and speed time graphs - Solving problems involving travel graphs	- Ticker timer - ICT tools - Relevant text

TOPIC 6: VARIATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Variation	- express inverse variation in algebraic term - distinguish between direct and inverse variation - sketch graphs to illustrate direct and inverse variation - solve problems involving variation	- Direct variation - Inverse variation	- Discussing relationships showing direct or inverse variation - Discussing examples of - Direct variation - Inverse variation - Sketching graphs of inverse functions - Solving problems involving variation	- ICT tools - Relevant text

TOPIC 7: ALGEBRA

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Algebraic Manipulation	- find HCF and LCM of algebraic expressions - simplify algebraic fractions - factorise quadratic expressions of the form $ax^2 + bx + c$ where $a > 1$	- LCM and HCF of algebraic expressions - Algebraic expressions	- Finding LCM and HCF of algebraic expressions - Simplifying algebraic fractions - Factorise quadratic expressions	- ICT Tools - Relevant text

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Equations	• solve linear simultaneous equations using: • elimination • substitution • graphical method • solve quadratic equations using: • factorisation • graphical methods • solve simultaneously linear and quadratic equations • change the subject of a non linear formula	• Simultaneous equations • Quadratic equations • Change of subject • Substitution of values	• Solving linear equations by • substitution • elimination • drawing graphs of • linear • quadratic equations using graphs • solving linear and quadratic equations using graphs • solving quadratic equations using • factorisation • graphical method • solving problems from life situations using equations • Discussing change of subject and its applications	• e-learning software on equations • internet
Inequalities	• solve simultaneous linear inequalities in one variable by calculation • represent solution set on line graph • solve simultaneously linear inequalities graphically	• Simultaneous inequalities • Graphs of inequalities	• Solving simultaneous linear inequalities in one variable • Representing solution set on a line graph • Representing linear inequalities in two variables on the Cartesian plane by shading the unwanted regions	• ICT Tools • Relevant text

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
			Representing the solution set of simultaneous linear inequalities in a Cartesian plane	
Indices and Logarithms	- simplify algebraic expressions involving indices - define logarithm - evaluate logarithms - state the laws of logarithms - simplify expressions using laws of logarithms - solve equations involving - indices - logarithm	- Indices - Logarithms	- Simplifying algebraic expressions involving indices - Discussing logarithms - Evaluating logarithms exploring and discovering laws of logarithms - Simplifying expressions using laws of logarithms - Solving equations involving - Indices - Logarithms	- ICT Tools - Relevant text

TOPIC 8: GEOMETRY

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Points, lines and angles	- construct angles of - elevation - depression - use scale drawings to solve problems on angles of elevation and depression	Angle of elevation and depression	- Constructing angles of elevation and depression - Solving problems on angle of elevation and depression using scale drawing	Local environment
Bearing	- illustrate bearing diagrammatically - use three figure bearing measured clockwise from north and compass bearing to solve problems	- Three figure bearing - Compass bearing	- Constructing a diagram to show bearing - Solving problems involving three figure bearing and compass bearing - Locating the position of an object using three figure bearing and compass bearing	- Models - ICT Tools - Compass - Maps
Polygons	- give properties of polygons - solve n-sided polygon - use the properties of: - radius - diameter - chord to solve problems	- Polygon - Interior angles - Exterior angles	- Stating properties of n-sided polygons - Solving problems involving n-sided polygons	- Models - ICT Tools

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Similarity and Congruency	- find the scale factor from two given similar shapes - calculate the area of similar figures - calculate the volume and mass of similar solids	- Areas of similar figures - Volume and mass of similar solids	- Discussing scale factor, area factor and volume factor - Computing areas of similar shapes - Solving problems on volumes and masses of similar solids	- Models of solids - Cuttings of plane shapes
Constructions	- construct triangles - construct quadrilaterals - solve life problems using construction of triangles and quadrilaterals	- Triangles - Quadrilaterals	- Constructing triangles and quadrilaterals - Solving problems using construction of triangles and quadrilaterals	ICT Tools
Symmetry	- State the order of rotational/point symmetry - Solve problems involving rotational symmetry	Rotational symmetry in two dimensions	- Discussing rotational/point symmetry - Constructing symmetrical plane shapes - Solving problems involving rotational symmetry in life	- Model shapes - ICT Tools

TOPIC 9: STATISTICS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Data Collection and Classification (grouped data)	- collect statistical data - group raw data into classes - state the class widths for the grouped data	- Data: - Collection - Classification - Class - width	- Gathering of statistical data - Conducting experiments to collect data - Classifying the collected data - Finding the class width of grouped data	- Scale - Tape measure - Local environment
Data Representation (grouped data)	- construct frequency tables - draw: - bar chart - pie chart - histogram - frequency polygon - analyse information on the graphs	- Frequency table - Bar chart - Pie chart - Histogram - Frequency polygon	- Drawing up frequency tables - Constructing graphs - Interpreting the graphs	- Flyers - Newspapers cuttings - ICT Tools
Measures of Central Tendency (grouped data)	- compute the mean - find the mode and median - use the assumed mean method to calculate the mean	- Mean, median and mode - Assumed mean	- Calculating the mean averages - Computing the mean using the assumed mean method - Finding the mode and the median of measure of central tendency - Explain the significance of measures of central tendency	- Flyers - Newspapers cuttings - ICT Tools

TOPIC 10: TRIGONOMETRY

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Pythagoras Theorem	- derive the Pythagoras theorem - solve right angled triangle - apply the Pythagoras theorem	Pythagoras theorem	- Using the method of counting squares to derive the Pythagoras theorem - Finding the missing side in right angled triangles using Pythagoras theorem - Solving problems in everyday life using the Pythagoras theorem	- Local environment - ICT Tools - Relevant text
Trigonometrical Ratios	- find sine, cosine, tangent of acute and obtuse angles - solve problems involving right angled triangles in two dimensions	- Trigonometrical ratios of acute and obtuse angles - Sine - Cosine - Tangent	- Calculating sine, cosine and tangent of acute and obtuse angles - Solving problems involving right angled triangles in two dimensions	- Local environment - ICT tools - Relevant text

TOPIC 11: VECTORS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Types of Vectors	- describe types of vectors - represent types of vectors on Cartesian plane - identify various types of vectors from the Cartesian plane	- Translational vectors - Equal vectors - Negative vectors - Parallel vectors - Position vectors	- Discussing various types of vectors including vector notation - Drawing different types of vectors on a Cartesian plane - Identifying different types of vectors on a Cartesian plane	- Geo board - ICT tools - Graphs - Relevant texts - Local environment
Operations	- add vectors - subtract vectors - multiply a vector by a scalar - calculate the magnitude of a vector - solve problems involving vector operations	- Vectors: - Addition - Subtraction - Scalar multiplication - Magnitude - Combined vector operations	- Manipulating vectors by: - Adding - Subtracting - Multiplying by a scalar - Computing the magnitude of a vector - Solving problems involving vectors	- Geo board - ICT Tools

TOPIC 12: MATRICES

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Dimension/ Order	•			
Operations	- add matrices - subtract matrices - multiply a matrix by a scalar - multiply matrices	- Addition and subtraction of matrices - Scalar multiplication of matrices - Multiplication of matrices	- Summing up matrices - Subtracting matrices - Using scalar quantities to multiply matrices - Identifying where matrix operations are possible - Multiplying matrices - Using matrices to solve problems	- ICT Tools - Local environment - Relevant text
Determinants	- find the determinant of a 2×2 matrix - distinguish between singular and non-singular matrices	- Determinant - Singular matrices - Non-singular matrices	- Calculating the determinant of 2×2 matrices - Differentiating between singular and non-singular matrices - Using the standard notation A or $\text{Det } A$	- ICT Tools - Relevant text
Inverse Matrix	- find the inverse of a 2×2 matrix - solve simultaneous equations using the matrix method	- Inverse - Simultaneous linear equations in 2 variables	- Calculating the inverse of a 2×2 non-singular matrix - Using the standard notation A^{-1}	- ICT Tools - Relevant text

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
			Solving simultaneous equations using the inverse of a 2×2 non singular matrix	

TOPIC 13: PROBABILITY

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Experimental and theoretical probability	- describe: - experimental probability - theoretical probability - deduce probabilities from results of experiments - identify situations where experimental or theoretical probabilities are used	- Experimental probability - Theoretical probability	- Discussing theoretical and experimental probability - Citing situations where experimental or theoretical probabilities are used - Carrying out experiments such as tossing a coin and throwing a die - Computing probabilities of events	- Print media - Spinning wheel - Coins - Dice - ICT Tools - Relevant text
Single events	- calculate probabilities of single events - use probability rules to compute probabilities of single events	- Single events - Probability - space - rules	- Carrying out experiments of single events - Computing probabilities of single events	- Coins - Dice - Balls - Playing cards - ICT Tools - Relevant text

TOPIC 14: TRANSFORMATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Translation	translate plane figures on Cartesian Plane using translation vectors	Translation vectors	- Drawing of plane shapes on the Cartesian Plane - Moving plane figures/shapes using translation vectors	- Geo-board - Cuttings of plane shapes/figures
Reflection	- reflect plane figures in a line of reflection - find the axis of reflection of given objects and images	- Reflection - Axis of reflection $x - axis$ $y - axis$ $y = a$ $x = b$	- Drawing images of plane figures under reflection - Finding coordinates of images of plane figures under reflection - Determining the axis of reflection	- Geo-board - Magazines - Mirrors - Local environment - ICT Tools
Rotation	- describe rotation - rotate points on a Cartesian plane by drawing - rotate plane figures on a Cartesian plane - find the centre of rotation by drawing - determine the angle of rotation by measuring	- Rotation - Perpendicular bisectors of lines - Direction: - Clockwise - Anticlockwise	- Explaining rotation - Drawing images and objects on the Cartesian plane - Describing a rotation completely given the objects and images on the Cartesian plane - Describing a rotation completely given the object and images *Notes: complete description of rotation consist of - centre of rotation - angle of rotation	- Geo board - Magazines - Local environment - ICT Tools

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
			Direction (clockwise and anticlockwise)	
Enlargement	- describe enlargement - enlarge plane figures about the origin using a rational scale by drawing - find the scale factor	- Enlargement - Scale factor	- Discussing enlargement - Drawing images of plane figures - Determining the scale factor (enlargement factor)	- Geo-board - Magazines - Local environment - ICT Tools

FORM FOUR (4)

8.0 COMPETENCY MATRIX

TOPIC 1: REAL NUMBERS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Approximation and Estimation	solve problems involving limits of accuracy	Limits of accuracy	Computing the minimum and maximum areas and perimeters of plane shapes	- ICT Tools - Relevant text

TOPIC 3: FINANCIAL MATHEMATICS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Consumer arithmetic	- identify currencies used in different countries - explain foreign exchange - convert from one currency to another - explain sales tax and income tax	- Foreign exchange - Sales and income tax rates (Pay as you Earn) PAYE - Value Added Tax (VAT) - Customs and excise duty	- Discussing foreign exchange and types of taxes - Calculating problems involving foreign exchange and types of taxes - Participating in educational tours	- Newspaper cuttings - ICT Tools - Till slips - Invoices - Receipts - Resource person - Pay slips

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- explain value added tax (VAT) - explain customs and excise duty - calculate Value Added Tax (VAT)			

TOPIC 4: MEASURES AND MENSURATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Mensuration	- identify similar shapes - calculate area and volume of similar shapes - calculate surface area - calculate volume of prisms and frustums	- Area and volume of similar shapes - Surface area - Volume of prisms - Volume of frustums	- Discussing similar shapes - Computing area and volumes of similar shapes - Calculating surface area - Computing volumes of prisms and frustums	- Models of shapes - Print media - Relevant text

TOPIC 5: FUNCTIONAL GRAPHS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Functional Graphs	- draw cubic graphs - draw inverse graphs - solve problems involving cubic or inverse functions	- Cubic functions - Inverse functions - Cubic graphs - Inverse graphs	- Discussing cubic and inverse functions - Drawing cubic graphs - Drawing graphs of inverse functions of the form $\frac{a}{bx+c}$ where a, b and c are integers - Solving problems involving cubic or inverse functions	- ICT Tools - Relevant text - Geo board
Travel Graphs	- explain the relationship of displacement, velocity, acceleration and time - draw displacement time graphs - draw velocity-time graphs - solve problems involving displacement-time and velocity-time graphs	- Displacement time graphs - Velocity-time graphs - Displacement - velocity - acceleration	- Discussing displacement velocity acceleration and time - Drawing: - Displacement-time graphs - Velocity-time graphs - Solving problems involving displacement-time and velocity-time graphs	- ICT Tools - Relevant text - Geo board - Local environment - Ticker timer

TOPIC 6: VARIATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Variation	- determine a connecting formula for - joint variation - partial variation - calculate unknown variable(s) using the formula(e) - construct graphs to show relationship between variables - solve problems involving variation	- Joint variation - Partial variation	- Discussing joint and partial variations - Computing unknown variables using the appropriate formula - Sketching variation graphs - Solving problems from life situations involving variation	- ICT Tools - Relevant text - Geo board - Local environment

TOPIC 7: ALGEBRA

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Algebraic Manipulation	simplify algebraic fractions	- Algebraic fractions - Factorisation	- Simplifying algebraic fractions using - LCM of denominators - factorisation	- ICT Tools - Relevant text - Local environment
Equations	- solve quadratic equations by completing the square - derive the quadratic formula - solve problems by applying the quadratic formula	- Completing the square - Quadratic formula	- Solving quadratic equations by completing the square - Deriving the quadratic formula by completing the square - Solving problems from life situations using the quadratic formula	- ICT Tools - Relevant text - Local environment
Inequalities	- express a given life situation in algebraic form - represent inequalities on the Cartesian plane - solve life problems using inequalities	Linear programming	- Discussing formulation of inequalities from given life situations - Representing inequalities on a Cartesian plane - Solving life problems using inequalities	- ICT Tools - Relevant text - Local environment

TOPIC 8: GEOMETRY

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Polygons	calculate angles using circle theorems	Circle theorems	- Finding angles using properties of a circle - Tangent - Cyclic quadrilaterals - Applying the properties of a circle to solve related problems - Finding angles using circle theorems associated with: - angle subtended at the centre and on the circumference - angle in a semi-circle - angle in the same segment - angle in the alternate segment - cyclic quadrilateral - tangent and radius	- Relevant text - ICT Tools
Constructions	- define locus - solve problems using loci and/or - Bearing	- Loci - Perpendicular line	- Discussing locus - Exploring ways of constructing given loci including those involving	- Relevant text - ICT Tools - Local environment

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- Scale • Angles of elevation and depression • Area • construct locus of points in a plane which are • equidistant from a fixed point • a fixed straight line • two fixed points • two intersecting lines		- Circle - Parallel lines - Bisecting angles - Bisecting straight lines • Solving life problems using loci and/or - Bearing - Scale - Angles of elevation and depression - Area • Dropping a perpendicular from a point to a given line	• Mathematical set

TOPIC 9: STATISTICS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Data Representation (grouped data)	- construct frequency tables - draw histograms, frequency polygon, and cumulative frequency curve	- Frequency table - Histogram - Frequency polygon - Cumulate frequency table - Cumulative frequency curve	- Constructing frequency tables - Drawing frequency polygons and histograms - Drawing the cumulative frequency curves	- Newspaper cuttings - ICT tools - Relevant text
Measures of Central Tendency	find the median from cumulative frequency curve	median	Determining the median from the cumulative frequency curve (ogive)	- ICT Tools - Relevant texts
Measures of Dispersion	- find the quartiles from cumulative frequency curve (ogive) - calculate the: - interquartile range - semi inter-quartile range	- Range - Quartiles - Interquartile range - Semi inter-quartile range	- Estimating the quartiles from cumulative frequency curve - Computing the inter-quartile range and semi inter-quartile range - Discussing the importance of interquartile and semi inter-quartile range	- ICT Tools - Magazines - Newspapers

TOPIC 10: TRIGONOMETRY

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Trigonometrical Ratios	- solve triangles using cosine and sine rule - use the cosine rule and sine rule to solve 3 dimensional problems	- cosine rule - sine rule	- Converting degrees to minutes and minutes to degrees - Using the cosine rule and sine rule to solve triangles - Solving 3 dimensional problems using the sine and cosine rule	- ICT Tools - Relevant text - Local environment

TOPIC 11: VECTORS

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Operation on plane shapes	- express edges and diagonals of plane shapes as linear combination of vectors - find numerical values of scalars in equal vectors - determine ratio of parallel edges/diagonals of plane shapes	- Properties on plane shapes - Parallel vectors - Equal vectors - Simultaneous equations - Ratio	- Sketching of plane shapes - Representing edges and diagonals of plane shapes as linear combination of vectors - Calculating numerical values of scalars using equal vectors - Computing ratio of parallel edges/diagonals of plane shapes	- Geo board - Cuttings of plane shapes

TOPIC 12: PROBABILITY

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Combined Events	- define with examples combined events - construct outcome tables and tree diagram - use probability rules in the computation of probabilities	- Combined events - Outcome tables - Tree diagrams - Probability rules	- Discussing combined events - Citing examples of combined events - Constructing outcome tables and tree diagrams - Computing probabilities using probabilities rules	- Dice - Coins - Playing cards - Newspaper cuttings - Balls - ICT tools - Relevant text

TOPIC 13: TRANSFORMATION

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Translation	- translate vector from an object and its image on a Cartesian plane - calculate coordinates of images - illustrate images on a Cartesian plane	Vector operations	- Using the determined translation vector to find images of other points - Drawing images	- Geo-board - Cuttings of plane shapes

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
Reflection	- reflect plane figures in any line of the form $y = mx + c$ - determine matrices for the reflection in x and y – axes $y = x$ $y = -x$	- Reflection in the line $y = mx + c$ - Matrices of reflection - Matrix multiplication	- Drawing images of objects - Determining the axes of reflection - Calculating coordinates of images	- Geo-boards - Local environment - ICT Tools - Relevant text
Rotation	- rotate plane shapes using matrices - find matrices of rotations about the origin through angles which are multiples of 90° - describe the rotation completely given: - a matrix - object and image	- Matrices of rotation - Matrix operation	- Calculating coordinates of images using matrices - Drawing images of plane shapes - Determining the matrices of rotations - Explaining fully the rotation given the matrix and or image	- Geo board - Compasses and protractors - ICT tools - Relevant texts - Local environment
Enlargement	- enlarge plane figures using matrices about the origin - enlarge plane figures about any point using a rational scale by drawing - describe fully an enlargement for a: - stated matrix - object and image	Enlargement matrices	- Calculating coordinates of images using matrices - Drawing images of plane figures on the Cartesian plane - Explaining fully the enlargement for a given: - Matrix - Object and image	- Geo boards - ICT tools - Cuttings of plane figures - Relevant text - Local environment
Stretch	- describe a stretch - calculate coordinates of the image given the matrices	- Stretch - one way - two way	- Discussing a stretch - Computing coordinates of images given the matrices	- Elastic materials - Geo boards

SUB TOPIC	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED NOTES AND ACTIVITIES	SUGGESTED RESOURCES
	- draw images of plane figures given matrices - describe fully a stretch given a - matrix - image	Invariant line and point	- Plotting images of plane figures given matrices - Identifying invariant line/point - Explaining a stretch fully given a matrix or image by stating: - Stretch factors - Invariant line	- Cuttings of plane figures - ICT Tools - Relevant text - Local environment
Shear	- describe a shear - compute coordinates of the images given a matrix - draw images of plane figures given the matrix - describe completely the shear given a - matrix - image	- Shear - Scale factor - Invariant line	- Discussing a shear - Calculating coordinates of image given the matrix - Plotting images of plane figures given matrices - Explaining fully a shear given a matrix or an image by stating an invariant line and shear factor	- Geo boards - Relevant text - ICT tools

9.0 ASSESSMENT

The Mathematics syllabus B learning area at Lower Secondary school module (Form 1-4) shall be assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA). These assessments shall be guided by the principles of inclusivity, practicability, authenticity, transparency, flexibility, validity and reliability. The principles are crucial for creating a supportive and effective learning environment that fosters growth and development in learners at secondary level. In addition to SBCA and SA those aspects that cannot be assessed through the continuous and summative modes will be assessed through learner profiling. The aspect to be profiled will include soft skills among others. Arrangements, accommodations and modifications must be visible to enable candidates with special needs to access assessments. This section covers the assessment objectives, the assessment model, the scheme of assessment, and the specifications for continuous assessment.

9.1 Assessment Objectives

By the end of the Mathematics syllabus B learning area for Forms 1- 4, learners will be assessed on their ability to:

- 9.1.1 recognise and use mathematical symbols, terms and definitions
- 9.1.2 apply mathematical symbols, terms and definitions
- 9.1.3 carry out calculations accurately
- 9.1.4 estimate to a suitable degree of accuracy
- 9.1.5 approximate to a suitable degree of accuracy
- 9.1.6 measure to a suitable degree of accuracy
- 9.1.7 draw tables, graphs, charts and diagrams accurately
- 9.1.8 read tables, graphs, charts and diagrams accurately
- 9.1.9 interpret tables, graphs, charts and diagrams accurately
- 9.1.10 carry out geometrical constructions and manipulations accurately
- 9.1.11 convert mathematical information from one form to another
- 9.1.12 deduce and draw inferences through manipulation of statistical data
- 9.1.13 solve routine and non-routine problems using appropriate formulae, algorithms and procedures

9.2 Assessment Model

Assessment of learners at Lower secondary school level for Mathematics syllabus B will be both Continuous and Summative as illustrated in Figure 1. Continuous Assessment will include recorded activities from the School Based Project and other activities done by the learners for assessment while summative will include terminal examinations, end of week, month, term, year or check points assessments.

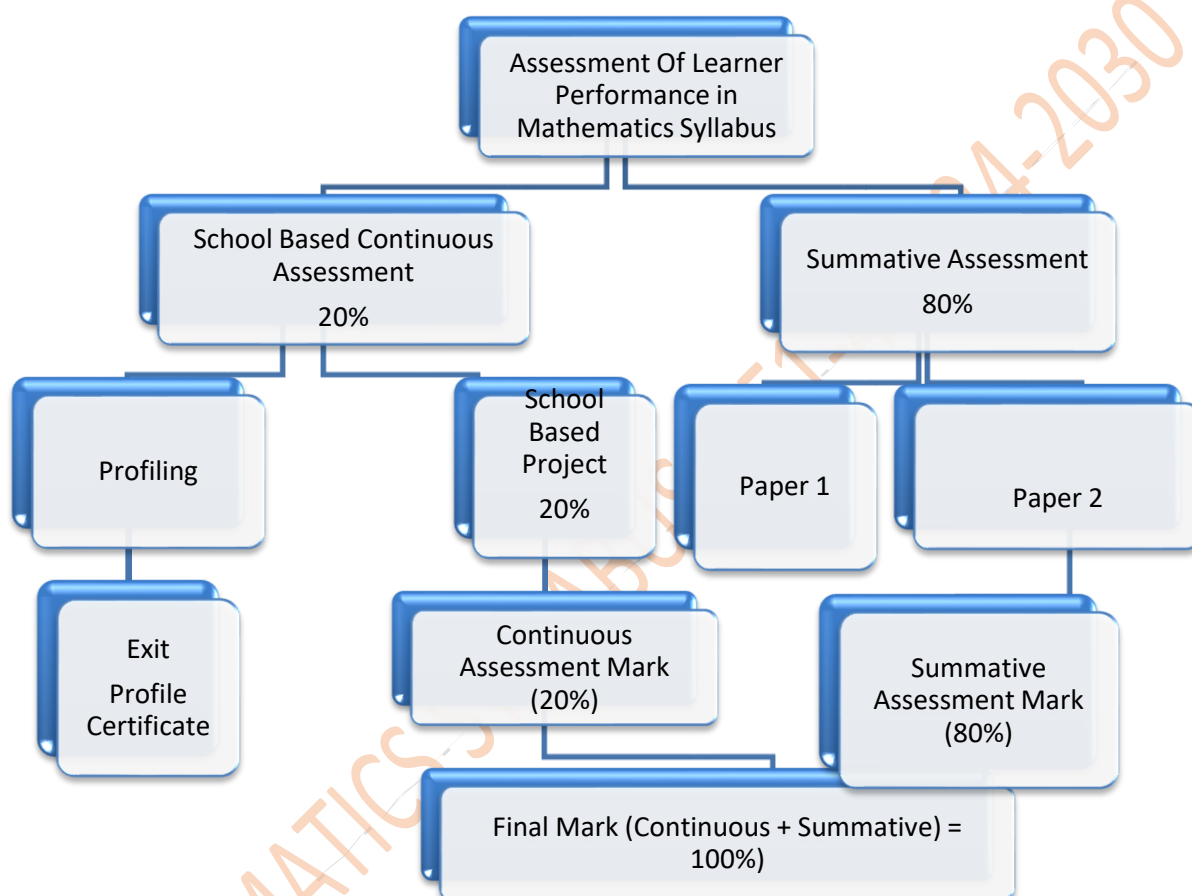


Fig. 1 Assessment Model

In addition, learners shall be profiled and learner profile records established. Learner profile certificates shall be issued for checkpoints assessment in schools as per the dictates of the Teacher's Guide to Learning and Assessment. The aspects to be profiled shall include learner's prior knowledge, values and skills, and subsequently the new competences acquired at any given point.

9.3 Scheme of Assessment

Learners should be assessed using both School Based Continuous Assessment and Summative Assessment.

FORM OF ASSESSMENT		WEIGHTING
School Based Continuous Assessment		20%
Summative Assessment		80%
Total		100%

Table ? shows the assessment criteria for the school based project.

9.3.1 Description of School Based Continuous Assessment

Learners should do one school-based project per form which contributes to 20% of the end of year final mark. However, two school based projects shall be considered for form 4 public examinations that is project done during their forms 3 and 4 sessions. Each will contribute 10%.

9.3.1.1: School – Based Project: Continuous Assessment Scheme

The table given below shows the Learning and Assessment Scheme for the school based project

Project Execution Stages	Project Stage Description	Timelines	Marks
1	Problem Identification	January	5
2	Investigation of related ideas to the problem/innovation	February	10
3	Generation of possible solutions	March	10
4	Selecting the most suitable solution	April-May	5
5	Refinement of selected solution	June	5
6	Presentation of the final solution	July	10
7	Evaluation of the solution and Recommendations	August-September	5
	TOTAL		50

The learning and assessment scheme shows the stages that shall be executed by pupils and the timeline at which each stage shall be carried out. Possible marks, totalling 50, are highlighted to indicate how much can be allocated.

9.3.2 Description of the ZIMSEC Summative Assessment

Summative assessment consists of two papers of equal weighting

Paper	Paper type	Marks	Duration	Weighting
1	Structured – short answer items	100	$2\frac{1}{2}$ hours	40%
2	Structured – short answer items and long questions	100	$2\frac{1}{2}$ hours	40%
TOTAL				80%

Paper 1

Duration: $2\frac{1}{2}$ hours

The paper consists of about 30 short structured compulsory question set on all syllabus topics and marked out of 100. The use of calculators in this paper is not allowed.

Paper 2

Duration: $2\frac{1}{2}$ hours

This paper consists of two sections A and B which will be set to cover all topics of the syllabus.

Section A: This section has five (5) compulsory questions, marked out of 52.

Section B: This section has seven (7) long questions. Candidates should answer four questions of their choice. Each question carries twelve (12) marks. The section is marked out of 48.

9.4 Specification Grid

Skill	Paper 1	Paper 2
Knowledge and comprehension	50%	50%
Application and Analysis	40%	40%
Problem solving	10%	10%
TOTAL	100%	100%